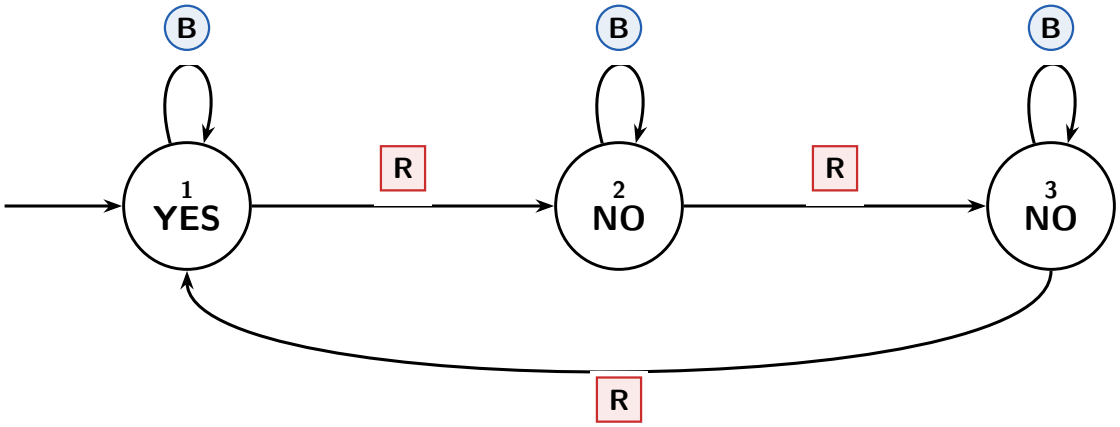
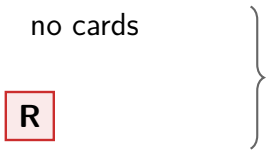
R means red; B means blue. A machine has one start circle and a fixed YES or NO on each circle. Each circle has exactly one R arrow and one B arrow. Read left to right with one marker, hiding used cards. Only the current circle and the card being read may affect the next move.

Problem 1: Find every five-card row this machine sends to YES. Describe exactly which rows it sends to YES, for any length.



Example R B: start 1 $\xrightarrow{R}$ 2 $\xrightarrow{B}$ 2; stop: NO.

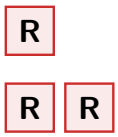
Problem 2: A row gets YES exactly when its red cards can be grouped in threes with none left over, including no red cards. For each pair, add the same extra cards in the same order to both rows to give one YES and one NO, or explain why this cannot be done. Adding no cards is allowed.



\+ _____

YES / NO

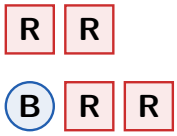
YES / NO



\+ _____

YES / NO

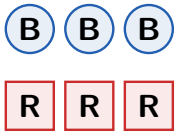
YES / NO



\+ _____

YES / NO

YES / NO



\+ _____

YES / NO

YES / NO

Problem 3: Two rows have left a machine on the same circle. Could adding the same extra cards in the same order to both ever give them different answers? Explain, for any machine obeying the rules on page 1.

Problem 4: Find the fewest circles needed by a machine that says YES exactly when its red cards can be grouped in fours with none left over. A row with no red cards must give YES. Draw a machine that works for rows of any length, and explain why no machine with fewer circles can work.

Problem 5: Find the fewest circles needed by a machine that says YES exactly when the last two cards are red followed by blue. Every other row must give NO. Draw a machine that works for rows of any length, and explain why no machine with fewer circles can work.

Problem 6: A machine must say YES exactly when its red cards can be grouped in fives with none left over. Find the fewest circles it needs, and do the same for groups of six. A row with no red cards must give YES. For any group size of one or more, describe a machine, explain why it works for every row, and explain why fewer circles cannot work.